

Project 3. Social Network Analysis

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Abstract—This report applies core social network analysis (SNA) techniques to the Spanish football transfer market for the 2025/2026 and 2026/2027 windows. A directed, weighted club-to-club network was mapped from 1,034 scraped transfer records covering LaLiga, LaLiga2, and five representative international leagues. We characterize the resulting 56-node network through its degree distribution, clustering, density, and shortest-path structure, visualize its community layout, and perform a link prediction analysis using four neighborhood-based similarity measures (Jaccard, Adamic-Adar, preferential attachment, and resource allocation), validated against held-out edges. The network is found to be small, dense, and highly clustered, with an average shortest path of 1.84 hops and a global clustering coefficient of 0.343. Link prediction achieves an estimated AUC of 0.71, confirming that transfer partners share structurally meaningful neighborhoods, though predictions concentrate around already well-connected hub clubs.

Index Terms—social network analysis, link prediction, degree distribution, football transfer market, NetworkX

I. INTRODUCTION

Football transfer markets are naturally representable as networks: clubs exchange players, and each transaction can be modeled as a directed edge from a selling club to a buying club. Beyond their competitive-sports interest, transfer networks display many properties studied in classic network science — hub dominance, tiered community structure (by division or financial power), and non-random link formation — making them a convenient real-world testbed for SNA techniques [1].

This project maps the transfer activity of the two top divisions of Spanish football (LaLiga and LaLiga2) for the current transfer cycle, and asks three questions: (i) what are the basic structural characteristics of this network, (ii) which clubs act as hubs or structural bridges, and (iii) can classic link prediction measures anticipate which club pairs are structurally likely to transact, even though the underlying score

is purely topological and ignores football-specific context such as budget or position needs.

II. MAPPING PROCESS

A. Data collection

The dataset was obtained through a purpose-built, rate-limited web scraper (`scraper.py`, built with `BeautifulSoup4`) that queries *Fichajes.com*, a public transfer-market tracker, for every club competing in Spain's First and Second Divisions. For each club, the scraper retrieves its arrivals and departures for the 2025/2026 and 2026/2027 seasons via the site's partial-content API. Each record stores the player's name and age, the transfer cost, the buying and selling clubs, and the season. Clubs outside Spain are consolidated into five representative nodes — *Premier League*, *Bundesliga*, *Ligue 1*, *Serie A*, and *Rest of the world* — to keep the graph at a manageable, interpretable scale, and reserve/youth ("B" or "II") teams are flagged as *filial* clubs. This process yielded **1,034** unique transfer records.

B. Graph construction

From the raw transfer list we built a **directed, weighted club-only graph** $G = (V, E)$: every club or league placeholder is a node, and a directed edge $u \rightarrow v$ is created whenever at least one player moved from club u to club v ; repeated transfers between the same pair are aggregated into a single edge whose weight equals the number of players moved. This yields $|V| = 56$ nodes and $|E| = 522$ directed (unique, aggregated) edges. For the structural measures that require an undirected simple graph (clustering coefficient, distances, link prediction), G is symmetrized into an undirected graph U with $|E_U| = 376$ edges. A secondary, heterogeneous club-player-club graph (856 nodes) was also built for an interactive

dashboard, but is not the focus of the structural analysis reported here.

C. Expectation before mapping

Before inspecting the data we expected the market to be dominated by a small number of high-activity hub clubs (big spenders and prolific sellers), and by close, tier-based clustering, i.e., clubs tend to trade more often within their own competitive tier.

III. BASIC CHARACTERISTICS AND VISUALIZATION

A. Global statistics

Table I summarizes the network’s core statistics, computed with *NetworkX* over the undirected simplification U .

TABLE I: Basic Network Characteristics

Metric	Value
Nodes	56
Edges (undirected)	376
Density	0.244
Average degree	13.43 (min 2, max 37)
Average clustering coefficient	0.392
Global clustering coeff. (transitivity)	0.343
Connected components	1
Diameter	3
Average shortest path length	1.84
Assortativity by division	0.006

The network is fully connected in a single component, has a small diameter of 3, and an average shortest path length under 2 — a signature "small-world" property: any two clubs are, on average, less than two intermediaries apart. The clustering coefficient (0.34–0.39) is substantially higher than what a random (Erdős–Rényi) graph of the same density would produce, indicating that clubs which share a trading partner are considerably more likely to trade with each other as well. The near-zero assortativity by division (0.006) was a mild surprise: we expected clubs to trade noticeably more within their own tier, but division alone barely predicts who trades with whom — other factors (loan pipelines, scouting networks, geography) appear to matter more than tier alignment.

B. Degree distribution

Figure 1 shows the degree histogram, and Figure 2 the same distribution on a log–log scale. The distribution is right-skewed: most clubs have a moderate degree (roughly between 8 and 16), while a small number of nodes — led by the *Rest of the world* placeholder (degree 37) and feeder-heavy clubs such as *Mirandés*, *Espanyol*, and *Alavés* (degrees 21–28) — act as hubs. The log–log plot shows a loosely heavy-tailed trend rather than a clean power law, which is expected given the small node count ($n = 56$); the qualitative hub-dominated pattern nonetheless matches our initial expectation.

C. Centrality

Table II lists the top-5 clubs by degree, betweenness centrality, and PageRank. The *Rest of the world* node dominates all three rankings, which is expected since it aggregates

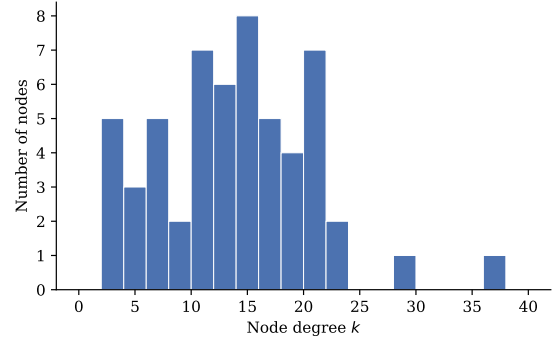


Fig. 1: Degree distribution of the club transfer network.

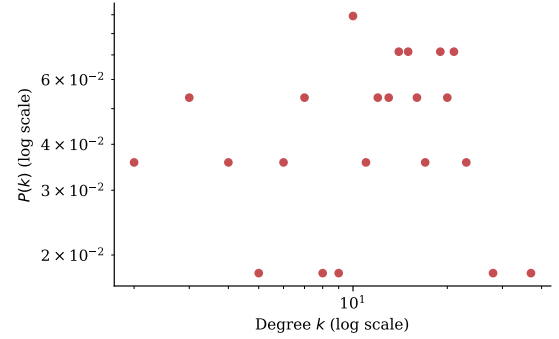


Fig. 2: Degree distribution on a log–log scale.

transactions with dozens of individual foreign clubs. More interestingly, among Spanish clubs the top bridges (highest betweenness) are not the biggest-spending clubs (Real Madrid, Barcelona, Atlético) but mid-table and lower-division sides with heavy squad rotation, such as *Mirandés* and *Alavés*. This is the clearest deviation from our initial expectation, and it illustrates a general SNA insight: raw transaction volume (degree) and structural bridging (betweenness) can single out different nodes as "important."

TABLE II: Top-5 Nodes by Centrality Measure

Degree	Betweenness	PageRank
Rest of the world (37)	Rest of the world (.113)	Rest of the world (.189)
Mirandés (28)	Mirandés (.086)	Premier League (.032)
Espanyol (23)	Alavés (.056)	Alavés (.029)
Alavés (23)	Espanyol (.046)	Mirandés (.029)
Elche (21)	Huesca (.037)	Espanyol (.028)

D. Visualization

Figure 3 shows a force-directed (spring) layout of the undirected network. Node color encodes competitive tier (international league, LaLiga, LaLiga2) and node size is proportional to betweenness centrality; only the highest-degree clubs are labeled to keep the figure readable. The visualization confirms the quantitative findings: a dense core of interconnected clubs surrounds a handful of oversized hub nodes, with international leagues sitting at the periphery but still tightly linked to the core through high-volume trading relationships.

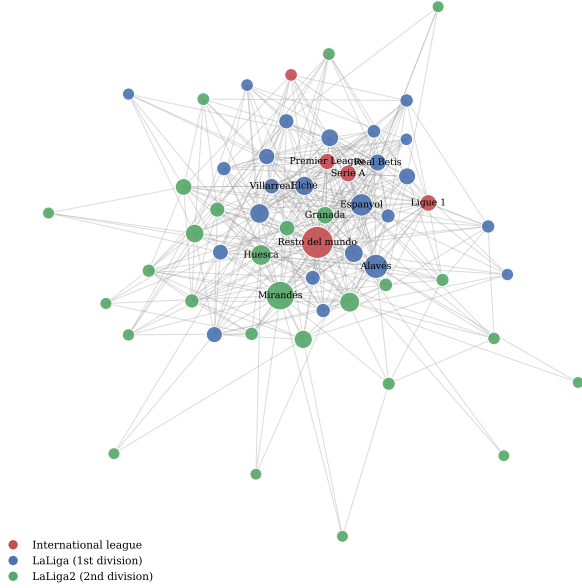


Fig. 3: Force-directed visualization of the LaLiga transfer market network. Node size $\propto$ betweenness centrality.

IV. LINK PREDICTION

A. Measures

We computed four classic neighborhood-based link-prediction scores over the undirected graph U , all provided by NetworkX: the Jaccard coefficient, the Adamic–Adar index, preferential attachment, and the resource allocation index. Each measure scores every currently non-adjacent pair of nodes; higher scores indicate a higher likelihood that a link should exist (or will appear) between the pair.

B. Expectation

Given the tiered structure observed above, we expected the top-ranked missing links to fall into two groups: (i) pairs of international-league placeholders, since these nodes already share almost all of their neighbors by construction, and (ii) pairs of clubs within the same division that had simply not yet transacted.

C. Results

Table III lists the top predicted pairs for each measure. As expected, international-league pairs (e.g. *Serie A–Rest of the world*, *Premier League–Ligue 1*) dominate every ranking, since these nodes are structurally near-duplicates of each other. Beyond the league placeholders, the measures increasingly diverge on which *club* pairs are prioritized: Jaccard and Adamic–Adar surface financially plausible, same-tier pairs such as *FC Barcelona–Real Madrid* and *Alavés–Real Betis*, while preferential attachment — being purely a function of raw degree — almost exclusively proposes pairs involving the highest-degree hub clubs (*Mirandés*, *Rest of the world*), regardless of whether they share meaningful common neighbors.

TABLE III: Top-3 Predicted Links per Measure

Measure	Top pair	Score
Jaccard	Premier League – Ligue 1	0.577
	Ligue 1 – Serie A	0.577
	FC Barcelona – Real Madrid	0.438
Adamic–Adar	Serie A – Rest of world	7.586
	Premier League – Rest of world	7.531
	Mirandés – Eldense	4.385
Pref. attachment	Premier League – Rest of world	777
	Serie A – Rest of world	777
	Ligue 1 – Rest of world	740
Resource alloc.	Serie A – Rest of world	1.332
	Ligue 1 – Rest of world	1.322
	Córdoba – Mirandés	0.797

D. Quantitative validation

Predicted links are only useful if the underlying measure has real discriminative power. To test this we hid 20% of the real edges (75 edges), recomputed the Adamic–Adar index on the remaining 80%, and evaluated two standard metrics against the hidden edges:

- **Precision@50:** of the top-50 ranked non-adjacent pairs, 11 (22.0%) were actual hidden edges, versus a 6.1% expected precision under random guessing — a 3.6 $\times$ improvement.
- **AUC estimate** (probability a random hidden edge out-ranks a random true non-edge): 0.71, comfortably above the 0.5 chance level.

E. Discussion

The validation confirms that neighborhood-based link prediction carries real signal on this network: an AUC of 0.71 means the shared-neighbor structure of the transfer market is meaningfully informative about which pairs of clubs are likely to trade. This matched our expectation that the measures would beat random guessing by a clear margin. What we did *not* fully anticipate is how strongly the small size and hub dominance of the network bias every measure toward re-proposing links between already highly-connected nodes, rather than surfacing novel, low-degree relationships; preferential attachment in particular reduces almost entirely to "connect the two biggest hubs." This suggests that, for a genuinely useful scouting application, topological scores like these should be combined with domain-specific features (financial capacity, positional needs, geographic proximity) rather than used in isolation.

V. CONCLUSION

The LaLiga transfer market forms a small-world, hub-dominated, moderately clustered network: 56 nodes reach each other in under two hops on average, with clustering well above the random-graph baseline. Centrality measures reveal that the most *active* clubs are not always the most *structurally important* bridges. Neighborhood-based link prediction, validated on held-out edges, achieves an AUC of 0.71, confirming genuine predictive signal, though its top recommendations skew toward

reinforcing existing hubs rather than discovering structurally novel trading relationships.

REFERENCES

- [1] A.-L. Barabási, *Network Science*. Cambridge University Press, 2018.